Routing, capacity and width in dynamic convolution: ResNet-18

Replication of the 132-run ResNet-18 program

Ankit Kumar
Mechatronics Department, University of Agder, Grimstad, Norway

Generated from result files by `report/build_model_report.py`

Built from the per-seed values printed in the ResNet-18 paper (test fixture).

This document reports one backbone of the cross-architecture replication. It is generated from the result files: every number, table and figure is computed, and every sentence states what a test returned. Interpretation across backbones is in the cross-model report. Method, statistical protocol and the pre-registered margin are those of the ResNet-18 study and are not repeated here.

1 Setup and coverage

Backbone. ResNet-18, adapted to 32×32 and 64×64 inputs as recorded in `src/models`. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$. Dynamic layers follow the ResNet-18 rule literally (every $k \times k$ convolution with $k > 1$ inside the blocks; stem, 1×1 convolutions, shortcuts, downsampling layers and head static): here the two 3×3 convolutions of every basic block, 16 layers. Initialisation: static convolutions keep PyTorch's default initialisation while every bank kernel is Kaiming-normal (fan-out), exactly as in the 132 original runs; this mismatch is disclosed in the ResNet-18 paper.

Size. At CIFAR-100 width 64 the static network stores 11.22 M parameters and the $K = 4$ `gap` network 44.46 M, of which 288.0 K are router parameters (Table 14). The bank adds 32.96 M to the static network; the router is 2.6 % of the static network's size. The router is the ResNet-18 design (a two-layer MLP with hidden width $C_{\text{in}}/4$ on the global descriptor), so its size grows with the square of a layer's input channels while a depthwise kernel grows linearly: where dynamic layers are depthwise, the routed arms carry many more parameters than the bank. The `gap−uniform` routing term includes them.

Gate. The first run, static, CIFAR-100, width 64, seed 0, reached 78.05 % against a pre-registered floor of 70.0 %: passed.

Coverage. 132 of 132 runs are present; 0 are missing, of which 0 diverged (non-finite training loss). 0 group(s) ran with gradient accumulation because batch 128 did not fit in memory. Table 1 lists every missing run and every deviation. Paired tests use the seeds both arms have; a verdict computed on fewer than six pairs carries its n.

Table 1: Missing runs, divergences and deviations from the recipe.

dataset	width	K	τ	arm	seed	status
none: all 132 runs present, batch 128 throughout, no resumed run						

2 CIFAR-100 at the standard width

Against `static`: `uniform` +0.048 pp (equivalent); `gap` +0.080 pp (equivalent); `shuffled` -0.073 pp (equivalent); `frozen_random` +0.177 pp (equivalent). 4 of the four arms are equivalent to `static` within 0.5 pp. The largest gap between arm means is 0.250 pp against a median within-arm seed standard deviation of 0.238 pp; one-way ANOVA across the five arms $F = 0.954$, $p = 0.450$. `frozen_random−gap` $= +0.097$ pp ($p = 0.443$, $p_{\text{TOST}} = 0.0089$). Power for `gap−uniform` here: paired sd 0.257 pp, minimum detectable difference 0.270 pp, simulated power 48.5, 96.3 and 100.0 % at true effects of 0.25, 0.5 and 1.0 pp. Table 2 gives the routing entropy by arm.

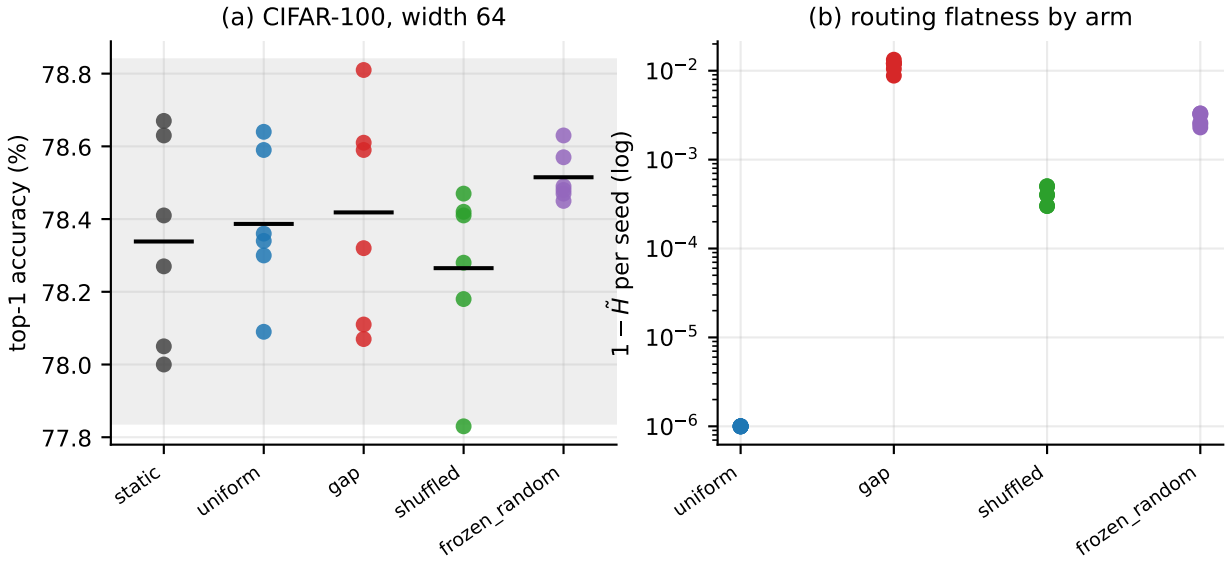


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

Table 2: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9885	0.9867	0.9912	0.01147
`shuffled`	0.9996	0.9995	0.9997	0.00040
`frozen_random`	0.9971	0.9967	0.9977	0.00287

3 The capacity sweep on CIFAR-100

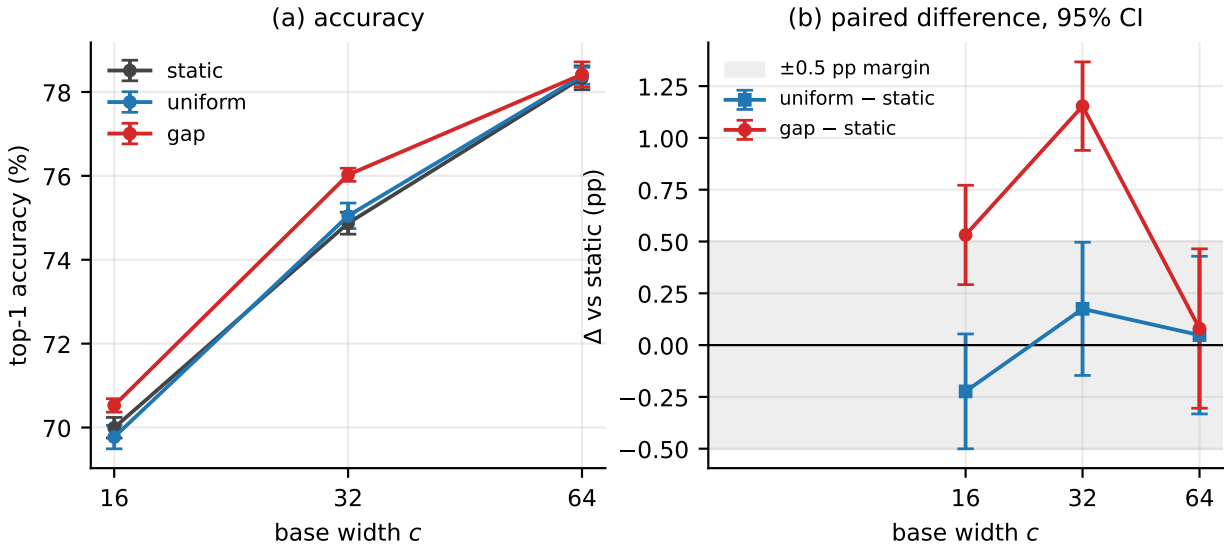


Figure 2: CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference against `static` with 95 % confidence intervals; the band is the ±0.5 pp margin.

`gap−static`: +0.532 pp at width 16 (higher), +1.153 pp at width 32 (higher), +0.080 pp at width 64 (equivalent) (Figure 2, Table 13).

4 Bank against routing

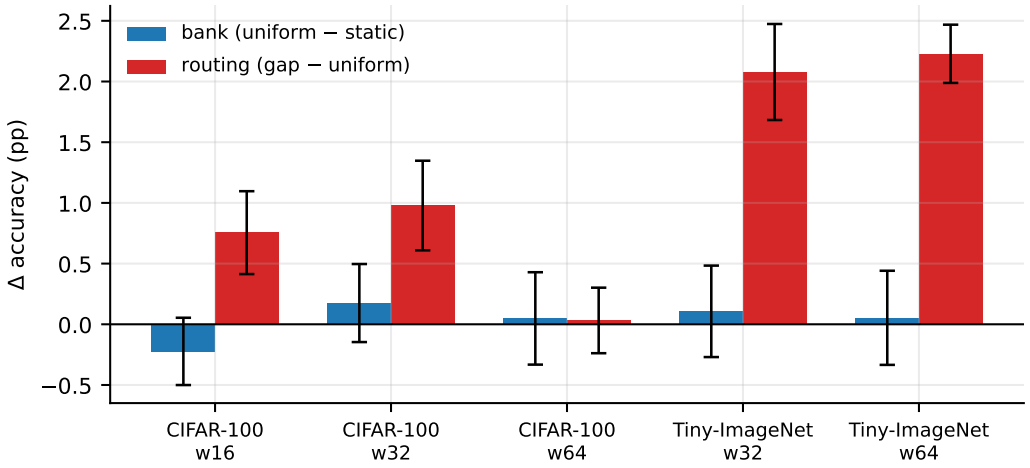


Figure 3: The total effect split into a bank term (`uniform−static`) and a routing term (`gap−uniform`), five conditions, 95 % confidence intervals.

The bank term is equivalent to zero in 5 of 5 conditions; the others: none. The routing term is positive with $p < 0.05$ in 4 of 5 conditions: CIFAR-100 width 16 +0.755 pp ($p = 0.0023$); CIFAR-100 width 32 +0.978 pp ($p = 0.0010$); CIFAR-100 width 64 +0.032 pp ($p = 0.7752$); Tiny-ImageNet width 32 +2.078 pp ($p = 0.0000$); Tiny-ImageNet width 64 +2.228 pp ($p = 0.0000$) (Table 3, Figure 3).

Table 3: Decomposition. Bank verdicts use TOST at ±0.5 pp.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	-0.223	0.0251	equivalent	+0.755	0.0023	+0.532	0.0023
CIFAR-100	32	+0.175	0.0242	equivalent	+0.978	0.0010	+1.153	$< 10^{-4}$
CIFAR-100	64	+0.048	0.0142	equivalent	+0.032	0.7752	+0.080	0.6158
Tiny-ImageNet	32	+0.107	0.0218	equivalent	+2.078	$< 10^{-4}$	+2.185	$< 10^{-4}$
Tiny-ImageNet	64	+0.053	0.0157	equivalent	+2.228	$< 10^{-4}$	+2.282	$< 10^{-4}$

5 Bank size

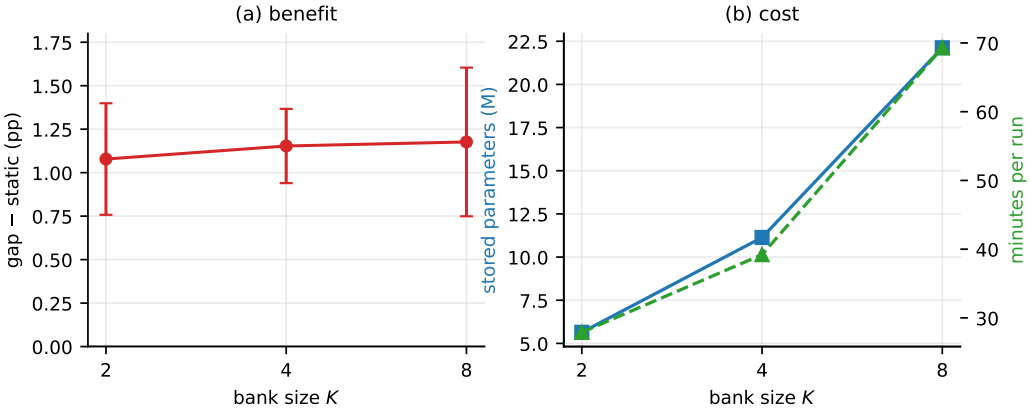


Figure 4: CIFAR-100 width 32, `gap` with $K = 2, 4, 8$. (a) Difference against `static`, 95 % CI. (b) Stored parameters and minutes per run.

$K = 8$ against $K = 2$: +0.098 pp (equivalent) for 3.92× the parameters (Tables 4, 5).

Table 4: Bank size at CIFAR-100 width 32. Differences are against `static`.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	75.950	0.148	+1.078	[+0.758, +1.399]	0.0003	5.64	27.9
4	76.025	0.156	+1.153	[+0.940, +1.367]	$< 10^{-4}$	11.13	39.2
8	76.048	0.242	+1.177	[+0.749, +1.604]	0.0009	22.12	69.3
`static`	74.872	0.262	–	–	–	2.82	8.6

Table 5: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.075	[-0.087, +0.237]	0.2878	0.0005	equivalent
$K = 8$ vs $K = 4$	+0.023	[-0.286, +0.332]	0.8538	0.0053	equivalent
$K = 8$ vs $K = 2$	+0.098	[-0.182, +0.379]	0.4091	0.0072	equivalent

6 Tiny-ImageNet

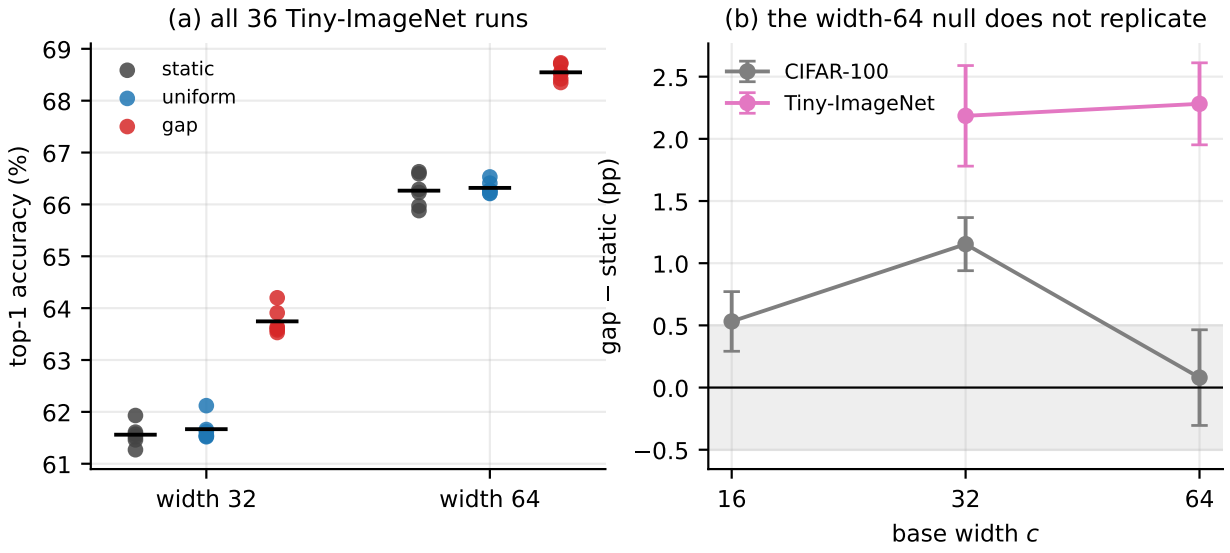


Figure 5: Tiny-ImageNet. (a) Every run at widths 32 and 64. (b) `gap−static` against width on both datasets, 95 % CI; the band is the margin.

`gap−static` is +2.185 pp at width 32 (higher) and +2.282 pp at width 64 (higher).

Across the five conditions the Spearman correlation between the static arm's final training loss and `gap−static` is $\rho = +0.30$ (Table 6, Figure 6).

Table 6: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap − static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0080	+0.080	[-0.305, +0.465]	equivalent
CIFAR-100	32	0.0170	+1.153	[+0.940, +1.367]	higher
Tiny-ImageNet	64	0.0210	+2.282	[+1.952, +2.611]	higher
Tiny-ImageNet	32	0.1194	+2.185	[+1.781, +2.589]	higher
CIFAR-100	16	0.1643	+0.532	[+0.292, +0.771]	higher

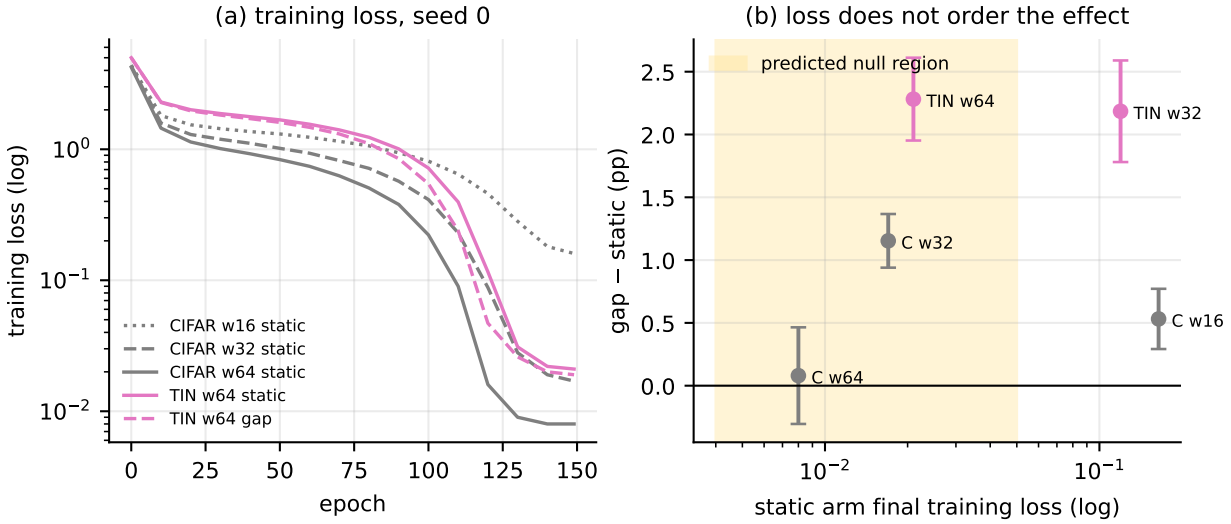


Figure 6: (a) Training loss, seed 0, sampled every ten epochs. (b) `gap−static` against the static arm's final training loss; the shaded region is where the ResNet-18 study predicted a null.

7 Equal parameter budgets

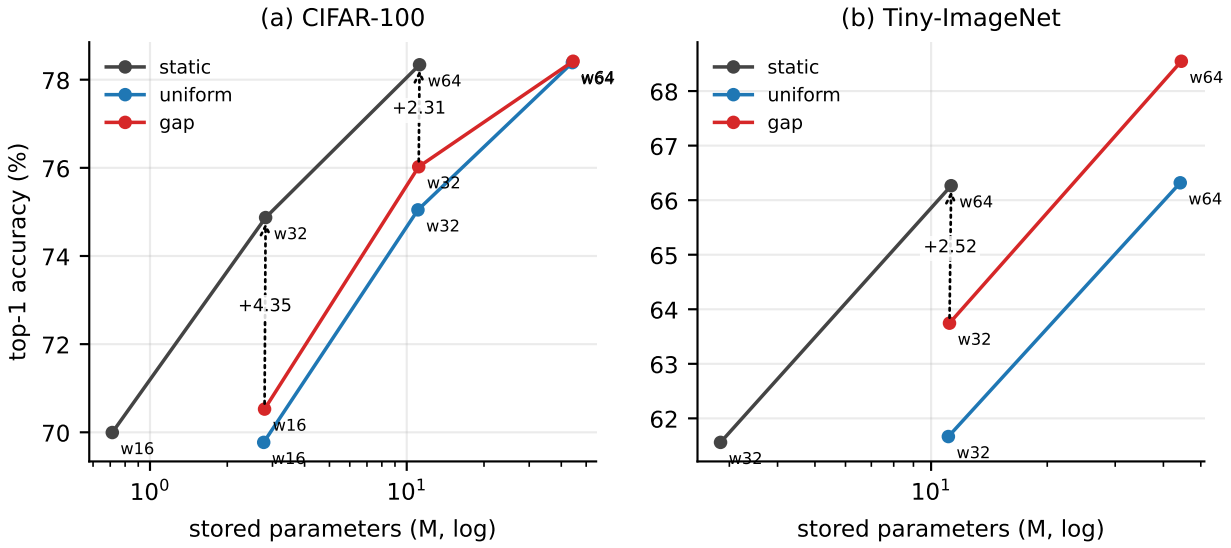


Figure 7: Accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; the label is the accuracy difference, and the parameter ratio where it is outside $\pm 10\,\%$.

The ResNet-18 comparison of `gap` at width c with `static` at width $2c$ was a matched-budget comparison because a $K = 4$ bank and a doubling of width both multiply the parameters of the dynamic layers by four. That holds only where almost every parameter is in a dynamic layer. Here 3 of 3 pairs are within 10 % and are called matched: CIFAR-100 `gap` w16 vs `static` w32: +4.345 pp, parameter ratio 0.99 (matched); CIFAR-100 `gap` w32 vs `static` w64: +2.313 pp, parameter ratio 0.99 (matched); Tiny-ImageNet `gap` w32 vs `static` w64: +2.520 pp, parameter ratio 0.99 (matched) (Table 7). Table 8 adds the comparison at equal parameters: the static accuracy curve interpolated linearly in log-parameters between the static widths that were run, evaluated at the `gap` network's count. It is an interpolation, not a run, and has no confidence interval.

Table 7: `gap` at width c against `static` at width $2c$. Δ is plain minus routed. The last column is the routed network's parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (2.79M)	`static` w32 (2.82M)	70.527	74.872	+4.345	[+4.103, +4.587]	$< 10^{-4}$	33.7 / 8.6	0.99 (matched)
CIFAR-100	`gap` w32 (11.13M)	`static` w64 (11.22M)	76.025	78.338	+2.313	[+1.870, +2.757]	$< 10^{-4}$	39.2 / 18.8	0.99 (matched)
Tiny-ImageNet	`gap` w32 (11.16M)	`static` w64 (11.27M)	63.745	66.265	+2.520	[+2.311, +2.729]	$< 10^{-4}$	284.2 / 129.0	0.99 (matched)

Table 8: `gap` against the static accuracy curve at equal parameters (interpolated, no CI).

dataset	routed	params (M)	routed acc.	static at equal params	Δ (pp)
CIFAR-100	`gap` w16	2.79	70.527	74.838	+4.311
CIFAR-100	`gap` w32	11.13	76.025	78.319	+2.294
Tiny-ImageNet	`gap` w32	11.16	63.745	66.231	+2.486

8 Temperature schedule

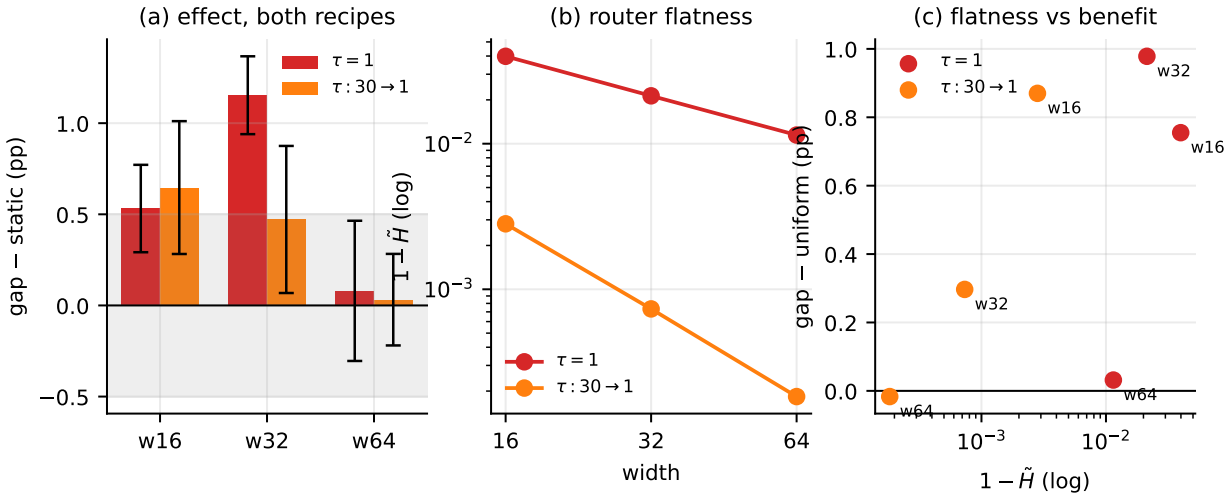


Figure 8: Constant $\tau = 1$ against τ annealed from 30 to 1 over ten epochs, CIFAR-100. (a) `gap−static`. (b) Router flatness. (c) Flatness against benefit over `uniform`.

Annealed minus constant, `gap`: width 16 +0.115 pp (equivalent); width 32 -0.682 pp (lower); width 64 -0.048 pp (equivalent) (Table 9).

Table 9: Temperature schedule, CIFAR-100.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	70.527	70.642	+0.115	0.4313	+0.532	+0.647	0.8259	+0.870	higher
32	76.025	75.343	-0.682	0.0017	+1.153	+0.472	0.4319	+0.297	inconclusive
64	78.418	78.370	-0.048	0.7449	+0.080	+0.032	0.0025	-0.017	equivalent

9 Routing entropy and routing benefit

At width 16 the constant-temperature router's $1 - \tilde{H}$ is 14.1 times the annealed router's; the benefit over `uniform` is +0.755 pp constant and +0.870 pp annealed. Across the 6 width–recipe points the Spearman correlation between $1 - \tilde{H}$ and `gap−uniform` is $\rho = +0.60$ (Table 10).

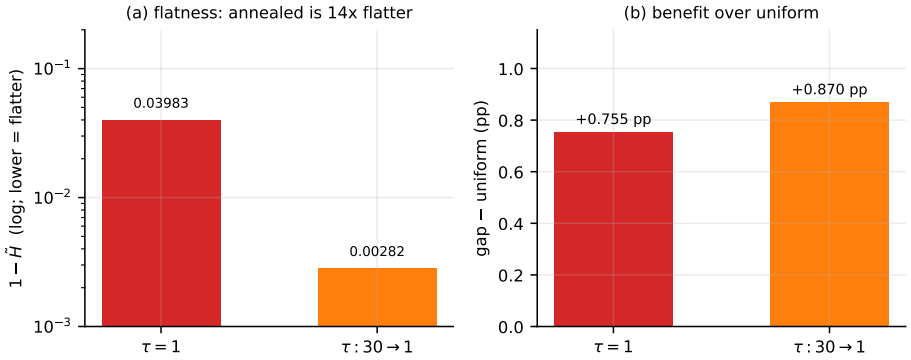


Figure 9: CIFAR-100 width 16, both temperature recipes. (a) Router flatness. (b) Benefit of `gap` over `uniform`.

Table 10: Routing entropy against the benefit over `uniform`.

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap`−`uniform` (pp)	p
16	1	0.96017	0.03983	0.0831	+0.755	0.0023
16	$30 \to 1$	0.99718	0.00282	0.0221	+0.870	0.0044
32	1	0.97868	0.02132	0.0608	+0.978	0.0010
32	$30 \to 1$	0.99927	0.00073	0.0113	+0.297	0.0810
64	1	0.98853	0.01147	0.0446	+0.032	0.7752
64	$30 \to 1$	0.99982	0.00018	0.0056	-0.017	0.8369

10 Pre-registered predictions

Recorded in writing, dated, before the Tiny-ImageNet width-64 runs of the ResNet-18 study.

Table 11: Pre-registered predictions and their outcomes.

	prediction	measured	outcome
(a)	`gap` − `static` below 0.5 pp, equivalent by TOST	+2.282 pp, $p = 1.0 \times 10^{-5}$, $p_{\text{TOST}} = 1.0000$	failed
(b)	static final training loss below 0.05	0.021	held

11 Complete results

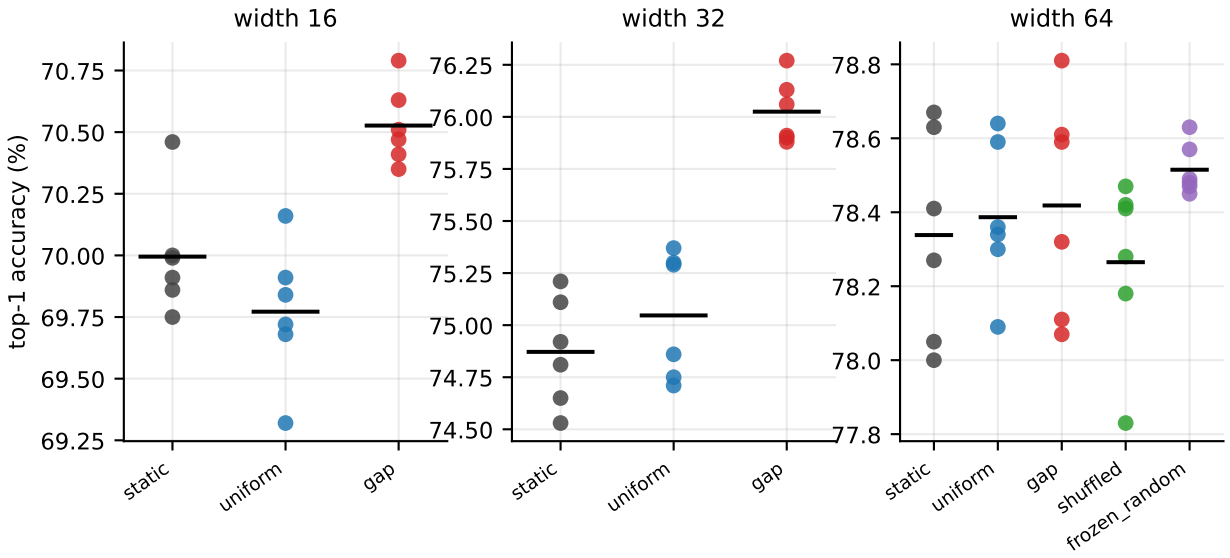


Figure 10: Every CIFAR-100 run at $K = 4$, $\tau = 1$. Bars are arm means.

Table 12: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). – marks a missing or diverged run (Table 1).

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	69.91	69.75	69.86	70.46	70.00	69.99	69.995	0.246
CIFAR-100	16	`uniform`	69.68	69.32	69.84	69.91	69.72	70.16	69.772	0.279
CIFAR-100	16	`gap`	70.35	70.51	70.41	70.63	70.79	70.47	70.527	0.160
CIFAR-100	32	`static`	74.81	74.65	74.53	74.92	75.11	75.21	74.872	0.262
CIFAR-100	32	`uniform`	75.29	74.86	74.75	75.30	74.71	75.37	75.047	0.305
CIFAR-100	32	`gap`	76.13	75.88	75.91	75.90	76.27	76.06	76.025	0.156
CIFAR-100	64	`static`	78.05	78.67	78.63	78.27	78.00	78.41	78.338	0.284
CIFAR-100	64	`uniform`	78.64	78.59	78.34	78.30	78.36	78.09	78.387	0.202
CIFAR-100	64	`gap`	78.59	78.81	78.11	78.61	78.07	78.32	78.418	0.299
CIFAR-100	64	`shuffled`	78.42	77.83	78.18	78.41	78.47	78.28	78.265	0.238
CIFAR-100	64	`frozen_random`	78.47	78.49	78.48	78.63	78.45	78.57	78.515	0.070
Tiny-ImageNet	32	`static`	61.93	61.61	61.57	61.46	61.27	61.52	61.560	0.217
Tiny-ImageNet	32	`uniform`	61.52	61.53	61.55	61.62	61.66	62.12	61.667	0.229
Tiny-ImageNet	32	`gap`	63.53	64.20	63.61	63.58	63.91	63.64	63.745	0.260
Tiny-ImageNet	64	`static`	66.29	66.59	65.88	65.97	66.63	66.23	66.265	0.309
Tiny-ImageNet	64	`uniform`	66.27	66.27	66.41	66.21	66.22	66.53	66.318	0.126
Tiny-ImageNet	64	`gap`	68.73	68.35	68.42	68.55	68.71	68.52	68.547	0.152

Table 13: Paired comparisons against `static`. p_W is the Wilcoxon signed-rank p (floor 0.031 at $n = 6$).

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	-0.223	[-0.500, +0.054]	0.0931	0.0938	-0.85	0.0251	equivalent
CIFAR-100	16	`gap`	+0.532	[+0.292, +0.771]	0.0023	0.0312	+2.33	0.6260	higher
CIFAR-100	32	`uniform`	+0.175	[-0.146, +0.496]	0.2206	0.3125	+0.57	0.0242	equivalent
CIFAR-100	32	`gap`	+1.153	[+0.940, +1.367]	$< 10^{-4}$	0.0312	+5.67	0.9997	higher
CIFAR-100	64	`uniform`	+0.048	[-0.332, +0.429]	0.7573	0.8438	+0.13	0.0142	equivalent
CIFAR-100	64	`gap`	+0.080	[-0.305, +0.465]	0.6158	0.5625	+0.22	0.0188	equivalent
CIFAR-100	64	`shuffled`	-0.073	[-0.602, +0.456]	0.7361	1.0000	-0.15	0.0464	equivalent
CIFAR-100	64	`frozen_random`	+0.177	[-0.121, +0.474]	0.1874	0.2188	+0.62	0.0191	equivalent
Tiny-ImageNet	32	`uniform`	+0.107	[-0.270, +0.483]	0.4994	0.6875	+0.30	0.0218	equivalent
Tiny-ImageNet	32	`gap`	+2.185	[+1.781, +2.589]	$< 10^{-4}$	0.0312	+5.67	0.9999	higher
Tiny-ImageNet	64	`uniform`	+0.053	[-0.334, +0.441]	0.7380	1.0000	+0.14	0.0157	equivalent
Tiny-ImageNet	64	`gap`	+2.282	[+1.952, +2.611]	$< 10^{-4}$	0.0312	+7.26	1.0000	higher

Table 14: Configuration and cost. Router parameters in thousands; training loss is the final-epoch mean over seeds; minutes are logged wall clock.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.71	0.0	0.1643	9.6
CIFAR-100	16	`uniform`	2.77	0.0	0.1653	11.4
CIFAR-100	16	`gap`	2.79	21.3	0.0638	33.7
CIFAR-100	32	`static`	2.82	0.0	0.0170	8.6
CIFAR-100	32	`uniform`	11.06	0.0	0.0170	12.9
CIFAR-100	32	`gap`	11.13	74.6	0.0132	39.2
CIFAR-100	64	`static`	11.22	0.0	0.0080	18.8
CIFAR-100	64	`uniform`	44.18	0.0	0.0081	25.2
CIFAR-100	64	`gap`	44.46	288.0	0.0074	86.0
CIFAR-100	64	`shuffled`	44.46	288.0	0.0083	92.3
CIFAR-100	64	`frozen_random`	44.18	0.0	0.0077	85.4
Tiny-ImageNet	32	`static`	2.85	0.0	0.1194	54.2
Tiny-ImageNet	32	`uniform`	11.09	0.0	0.1200	56.3
Tiny-ImageNet	32	`gap`	11.16	74.6	0.0420	284.2
Tiny-ImageNet	64	`static`	11.27	0.0	0.0210	129.0
Tiny-ImageNet	64	`uniform`	44.23	0.0	0.0220	141.0
Tiny-ImageNet	64	`gap`	44.52	288.0	0.0190	687.2

Table 15: Every run of this backbone. GPU-hours from logged wall clock.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	5.5
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	6.1
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	30.8
CIFAR-100	32	2	constant	`gap`	6	2.8
CIFAR-100	32	8	constant	`gap`	6	6.9
CIFAR-100	16	4	annealed	`gap`	6	4.4
CIFAR-100	32	4	annealed	`gap`	6	3.8
CIFAR-100	64	4	annealed	`gap`	6	8.6
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	39.5
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	95.7
total					132	204.1